

Weekly Self-Study Tasks to Complete

React Beyond the Basics for Leaders

This document covers all of Week 5 of Phase 2 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Page lab, where you will load data with `useEffect` and handle loading and error states. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep the DevTools Network tab open so you can see requests and loading states as the videos describe them.

Topic	Material Description	Est. Time	Link / Reference
Effects & Data Fetching	React Essential Training (Eve Porcello, LinkedIn Learning, released Oct 4, 2024; 1h 45m course). Watch the “React Server Components” and “Asynchronous React in Next.js” chapters (~25 min). Shows how modern React loads data and renders on the server. Reinforces the instructor demo.	~25 min	linkedin.com/learning – React Essential Training
useEffect	React.dev “Synchronizing with Effects” and “You Might Not Need an Effect” (official docs) — what effects are, when they run, and cleanup. Read and run the examples.	~20 min	react.dev/learn – Synchronizing with Effects

Topic	Material Description	Est. Time	Link / Reference
Routing	React Router “Tutorial” (official) — read and skim the first sections on routes, links, and URL params.	~15 min	reactrouter.com — Tutorial

Practice Activities (1 hour)

Complete this week’s hands-on lab: “Evergreen Insurance Quote Page — Loading Data with useEffect.” You will fetch the recent quotes from the server, handle the loading and error states, and clean up the request on unmount. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-react-data folder, run npm install, then npm run dev.
- Review the three state variables: quotes, loading, and error.
- Write a useEffect that fetches /quotes.json with an AbortController and an empty dependency array.
- Render the loading message, the error message, and the success list as separate states.
- Rename the data file to simulate an outage and watch the error state appear.
- Open the Network tab and notice the Strict Mode double-fetch in development.

The lab instructions (with “Why this matters” callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. Phase 3 turns to the back end (APIs, Node, databases) that these screens talk to, so focus this week on effects, data fetching, and routing.

Topic	Material Description	Link / Reference
React	Advanced React / React: Beyond the Basics (LinkedIn Learning) — a complete course covering effects, routing, forms, and performance in depth (Video Course).	linkedin.com/learning — Advanced React
Reference	React.dev “Escape Hatches” — the official section on effects, refs, and synchronizing with external systems (Reference / Tutorial).	react.dev/learn/escape-hatches
Routing	React Router documentation — the authoritative guide to client-side routing, params, and navigation (Reference).	reactrouter.com
Data Fetching	TanStack Query (React Query) Overview — the popular library many teams use instead of raw useEffect for server data (Reference).	tanstack.com/query
Performance	Keeping Components Pure & memo — React.dev. When and why to use React.memo, useMemo, and useCallback (Reference).	react.dev/reference/react/memo